1 Social Optimum: Braess Network

1.1 Network

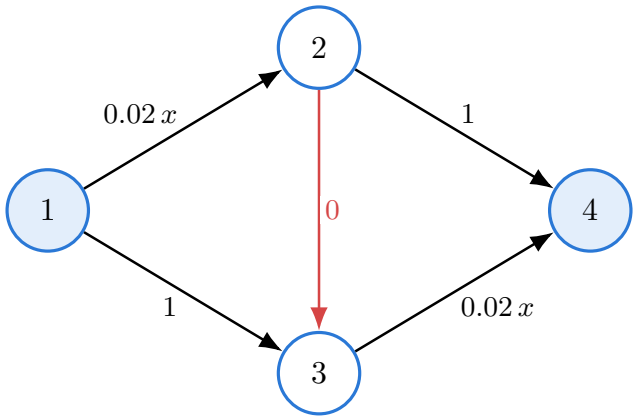


Figure 1: Braess network. x is the number of agents on an edge in a timestep; $2 \to 3$ is free.

All agents travel $1 \to 4$ on one of three paths, with $u + \ell + z = N$:

$$U = 1 \to 2 \to 4 \ (u), \qquad L = 1 \to 3 \to 4 \ (\ell), \qquad Z = 1 \to 2 \to 3 \to 4 \ (z).$$

1.2 Derivation

Timing. L agents cross $3 \to 4$ at step 1 and Z agents at step 2, so they never congest each other there. In the static model they would.

Cost. Edge $1 \to 2$ carries $N - \ell$ agents (step 0), and $3 \to 4$ carries ℓ (step 1) then z (step 2). Agents on U and L also pay 1. So

$$SC = 0.02\big[(N-\ell)^2 + \ell^2 + z^2\big] + (N - z), \qquad u = N - \ell - z \geq 0.$$

Optimum. The ℓ and z terms are independent, minimised at $\ell = N/2$ and $z = 25$.

- $N \geq 50$: then $u = N/2 - 25 \geq 0$, so this is optimal, and $SC_{\text{opt}} = 0.01N^2 + N - 12.5$.
- $N < 50$: u would be negative, so set $u = 0$ ($\ell = N - z$) and minimise again: $z = (N + 25)/3$. At $N = 20$, $z = 15$ and $\ell = 5$.

All values are whole numbers, so they are also the optimum for whole agents.

1.3 Results

N	u	ℓ	z	SC_{opt}	SC_{opt}/N	Static optimum/N
20	0	5	15	14.5	0.725	0.8
50	0	25	25	62.5	1.25	1.5
80	15	40	25	131.5	1.64375	1.8

Table 1: Social optimum in the simulator, with the classical static optimum for comparison.

- Static model: Z shares $3 \to 4$ with L, so the zig-zag pays only under light congestion. Everyone takes Z at $N = 20$; nobody does at $N = 50$ or 80.
- Simulator: Z reaches $3 \to 4$ a step after L, so the zig-zag always helps. For $N \geq 50$ it saves 12.5; at $N = 20$ it saves 1.5.
- $N = 50$ is the textbook Braess network. Selfish routing (everyone on Z) costs 100 against a static optimum of 75. The simulator's optimum, 62.5, is lower still.